

A Zero-Counting Obstruction for the Rank-One Shift Model in arXiv:1409.8569

Run fa_banach_001, agent agent_lane_03

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Status

Status: partial result, likely valid. This does not solve the problem of determining $q_B(p)$, but it proves that the rank-one ℓ_1 shift construction used in [1] cannot produce divergent eigenvalue-gap sums at any exponent $q > 1$.

Source Problem

Demuth, Hanauska, Hansmann, and Katriel study compact perturbations $L = L_0 + K$ of bounded operators on Banach spaces. If $(\alpha_j(K)) \in \ell^p$, their Corollary 1 implies, for $L_0 \neq 0$, the gap summability

$$\sum_{\lambda \in \sigma_d(L), |\lambda| > \|L_0\|} (|\lambda| - \|L_0\|)^q < \infty \quad (q > p + 1).$$

They define $q_B(p)$ as the infimum of exponents for which the corresponding implication holds for all Banach spaces, all L_0 , and all compact K with $(\alpha_j(K)) \in \ell^p$. Their results give $\max(1, p) \leq q_B(p) \leq p + 1$, and the final page asks whether $q_B(p) = p$ for $p \geq 1$, whether $q_B(p) > p$, or whether the upper bound $p + 1$ is sharp.

From (5.6) we thus obtain that

$$\sum_{\lambda \in \sigma_d(L), |\lambda| > \|L_0\|} (|\lambda| - \|L_0\|) = \sum_{w \in \mathbb{D}, h(w)=0} \frac{1 - |w|}{|w|} = \infty.$$

We thus conclude this article with a number of open problems: Is it true that for general Banach spaces we have that $q_B(p) = p$ for $p \geq 1$ (but maybe it will not be a minimum)? Or do we have that $q_B(p)$ is strictly larger than p ? Is our upper bound $p + 1$ actually equal to $q_B(p)$? In addition, one might also ask about the optimal exponents for more specific classes of Banach spaces (only the case of Hilbert spaces being known). These are, we believe, intriguing questions for further investigation.

Partial Result

The source paper obtains the lower bound $q_B(p) \geq 1$ from a rank-one perturbation of the unilateral shift on $\ell^1(\mathbb{N})$. In that model, a bounded scalar sequence $b = (b_k)_{k \geq 1}$ defines

$$h(w) = 1 - \sum_{k=1}^{\infty} b_k w^k, \quad |w| < 1,$$

and the eigenvalues λ of the perturbed shift outside the unit disk correspond to zeros $w = \lambda^{-1}$ of h in the unit disk. Formula (5.6) in [1] identifies the gap sum with

$$\sum_{\lambda \in \sigma_d(L), |\lambda| > 1} (|\lambda| - 1) = \sum_{w \in \mathbb{D}, h(w)=0} \frac{1 - |w|}{|w|}.$$

Theorem 1 (rank-one shift zero summability). *Let (b_k) be bounded and let $h(w) = 1 - \sum_{k \geq 1} b_k w^k$ in $\mathbb{D}$. If (z_n) is the zero sequence of h in $\mathbb{D}$, counted with multiplicity, then*

$$\sum_n (1 - |z_n|)^q < \infty \quad \text{for every } q > 1.$$

Corollary 1. *For every operator in the rank-one ℓ^1 shift model of [1, Section 5], the eigenvalues outside the unit disk satisfy*

$$\sum_{\lambda \in \sigma_d(L), |\lambda| > 1} (|\lambda| - 1)^q < \infty \quad \text{for every } q > 1.$$

Consequently this model cannot prove $q_B(p) > 1$, and hence cannot prove $q_B(p) > p$ for any $p \geq 1$ or sharpness of the upper bound $p + 1$.

Proof Intuition

The rank-one example is equivalent to prescribing zeros of an analytic function whose Taylor coefficients are bounded. Such functions may have non-Blaschke zero sets, which is exactly what the source paper uses to force divergence of the first power. But bounded coefficients impose the growth condition $|h(z)| = O((1 - |z|)^{-1})$. Jensen's formula converts that mild growth into a dyadic zero-counting estimate: near radius $1 - 2^{-m}$ there can be at most order $m2^m$ zeros. That count is just too small to make any q -power with $q > 1$ diverge.

Proof

Set $M = \sup_k |b_k|$. If $M = 0$, then $h \equiv 1$ and there are no zeros, so assume $M > 0$. For $0 < r < 1$,

$$|h(re^{it})| \leq 1 + M \sum_{k=1}^{\infty} r^k = 1 + \frac{Mr}{1-r} \leq \frac{1+M}{1-r}.$$

Since $h(0) = 1$, Jensen's formula gives, for every $0 < r < 1$,

$$\sum_{|z_n| < r} \log \frac{r}{|z_n|} = \frac{1}{2\pi} \int_0^{2\pi} \log |h(re^{it})| dt \leq C + \log \frac{1}{1-r},$$

where $C = \log(1 + M)$.

Let $N(t) = \#\{n : |z_n| \leq t\}$, counting multiplicity. For each integer $m \geq 1$, put $t_m = 1 - 2^{-m}$ and $r_m = 1 - 2^{-(m+1)}$. Then $t_m < r_m$, and each zero with $|z_n| \leq t_m$ contributes at least

$$\log \frac{r_m}{t_m} = \log \left(\frac{1 - 2^{-(m+1)}}{1 - 2^{-m}} \right) \geq 2^{-(m+1)}$$

to Jensen's sum at radius r_m . Hence

$$N(t_m) 2^{-(m+1)} \leq C + \log \frac{1}{1 - r_m} = C + (m + 1) \log 2.$$

Thus $N(t_m) \leq C_1 m 2^m$ for a constant C_1 independent of m .

Now decompose the disk into dyadic annuli

$$A_m = \{z : 1 - 2^{-m} < |z| \leq 1 - 2^{-(m+1)}\} \quad (m \geq 1),$$

together with the compact disk $|z| \leq 1/2$. The compact disk contains only finitely many zeros because h is not identically zero. The number of zeros in A_m is at most $N(t_{m+1}) \leq C_2 m 2^m$, while every $z \in A_m$ satisfies $1 - |z| \leq 2^{-m}$. Therefore

$$\sum_n (1 - |z_n|)^q \leq C_3 + \sum_{m=1}^{\infty} C_2 m 2^m 2^{-mq} = C_3 + C_2 \sum_{m=1}^{\infty} m 2^{-m(q-1)} < \infty$$

whenever $q > 1$. This proves the theorem.

For the corollary, the source model has $w = \lambda^{-1}$ and $h(w) = 0$. There are only finitely many zeros with $|w| \leq 1/2$. For the remaining zeros,

$$|\lambda| - 1 = \frac{1}{|w|} - 1 = \frac{1 - |w|}{|w|} \leq 2(1 - |w|),$$

so the theorem implies the asserted q -summability of eigenvalue gaps.

Limitations

This packet does not determine $q_B(p)$ and does not rule out other Banach spaces, other compact perturbations, or more elaborate finite-rank or infinite-rank constructions. It only shows that the natural attempt to strengthen the source paper's own rank-one ℓ^1 shift example cannot cross the exponent 1 threshold.

Novelty and Duplicate Check

The run indexes were searched on 2026-07-03 for arXiv:1409.8569, the paper title, $q_B(p)$, eigenvalue-gap keywords, and the authors' names; no prior packet or attempt for this target was found. Local parsed arXiv sources were searched for exact q_B and title phrases, again finding no later paper in the local corpus that resolves the question. External web searches on the same date for exact phrases including $q_B(p)$, the title, and the authors surfaced the original arXiv page but no separate later answer. This is an elementary partial obstruction, not a claimed complete or novel resolution of the open problem.

Human Review Recommendation

Review the Jensen-to-annular-counting estimate and the transfer from zeros w to eigenvalue gaps $\lambda = w^{-1}$. If those two steps are accepted, the partial obstruction is complete.

References

- [1] M. Demuth, F. Hanauska, M. Hansmann, and G. Katriel, *Estimating the number of eigenvalues of linear operators on Banach spaces*, arXiv:1409.8569, 2014.
- [2] J. B. Conway, *Functions of One Complex Variable*, 2nd ed., Graduate Texts in Mathematics 11, Springer, 1978.